13 A factory makes 450 pies every day.
The pies are chicken pies or steak pies.

Each day Milo takes a sample of 15 pies to check.

The proportion of the pies in his sample that are chicken is the same as the proportion of the pies made that day that are chicken.

On Monday Milo calculated that he needed exactly 4 chicken pies in his sample.

(a) Work out the total number of chicken pies that were made on Monday.

C = chicken pies; All = total pies.

	C	All	
×30	4	15	×30
	120	450	

1 mark
Scale the sample up by 450 ÷ 15 = 30

1 mark
4 × 30 = 120 chicken pies

120

(2)

On Tuesday, the number of steak pies Milo needs in his sample is 6 correct to the nearest whole number.

Milo takes at random a pie from the 450 pies made on Tuesday.

(b) Work out the lower bound of the probability that the pie is a steak pie.

Rounding to 6 gives 5.5 ≤ sample amount < 6.5.

1 mark
Use the lower bound, 5.5

Lower probability = $\frac{5.5}{15} = \frac{165}{450} = \frac{11}{30}$

1 mark
Correct lower probability

S = steak pies.

	S	All	
×30	5.5	15	×30
	165	450	

$\frac{11}{30}$

(2)

(Total for Question 13 is 4 marks)